PROJECT DOCUMENTATION



Deploying a Linux Web Server on AWS EC2

A beginner-friendly, step-by-step guide to hosting a web page with Apache on Amazon Linux 2023.

Security Groups | EC2 t2.micro | Amazon Linux 2023 | Apache HTTPD

YOUR JOURNEY

1. Security Group
2. Launch EC2
3. Connect
4. Install Apache
5. Access Page
6. Clean Up

CREATED BY

Umesh Kumbharkar

linkedin.com/in/umesh-kumbharkar

TOPIC

AWS EC2 Web Hosting

AWS EC2 | Apache | Linux

INTRODUCTION

Project Overview

This project guides you through deploying a **Linux-based web server on AWS EC2**. By the end of this tutorial you will have a fully functional web server running on **Amazon Linux**, capable of serving a basic HTML page. This setup is ideal for beginners looking to explore AWS cloud services and web hosting.



BEFORE YOU START

Prerequisites

- **An AWS account** with access to the EC2 service
- **Basic knowledge** of Linux commands and AWS Console
- **Internet access** a stable internet connection
- **A web browser** to open the AWS Management Console

THE PLAN

Project Roadmap

Security Group 01 Create the firewall rules PHASE 1	**Launch EC2** 02 Start the t2.micro server PHASE 2	**Connect** 03 Open the browser terminal PHASE 3
Install Apache 04 Set up the web server PHASE 4	**Access Page** 05 View it in a browser PHASE 5	**Clean Up** 06 Terminate the instance PHASE 6

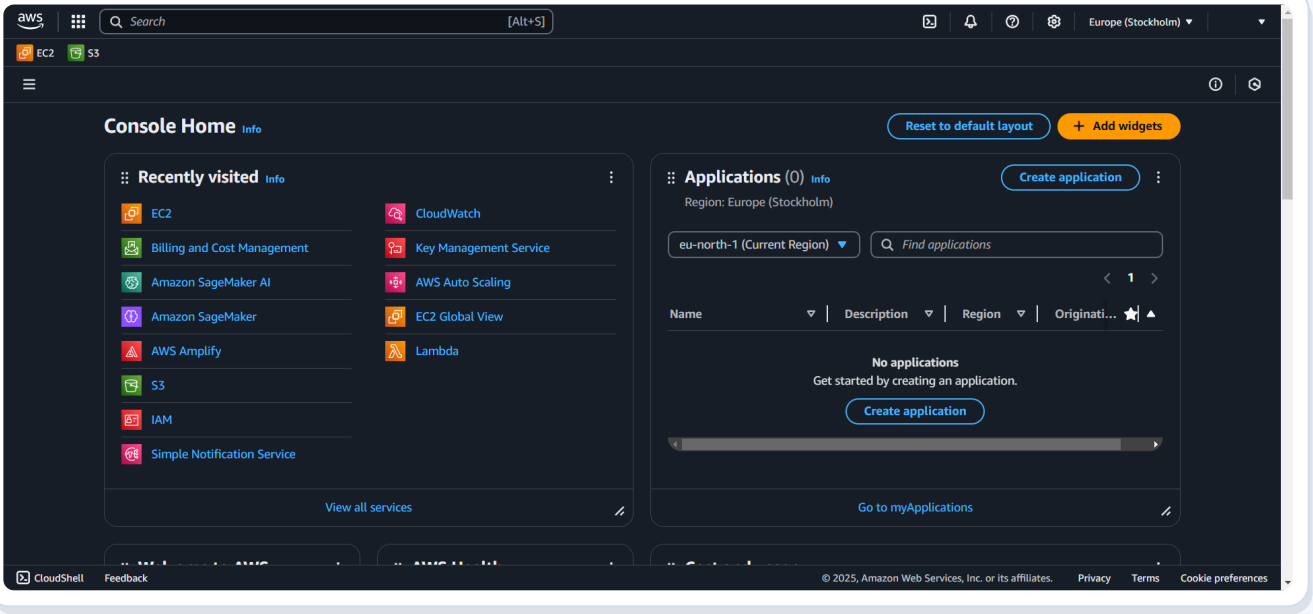
PHASE 1

Setting Up the Security Group

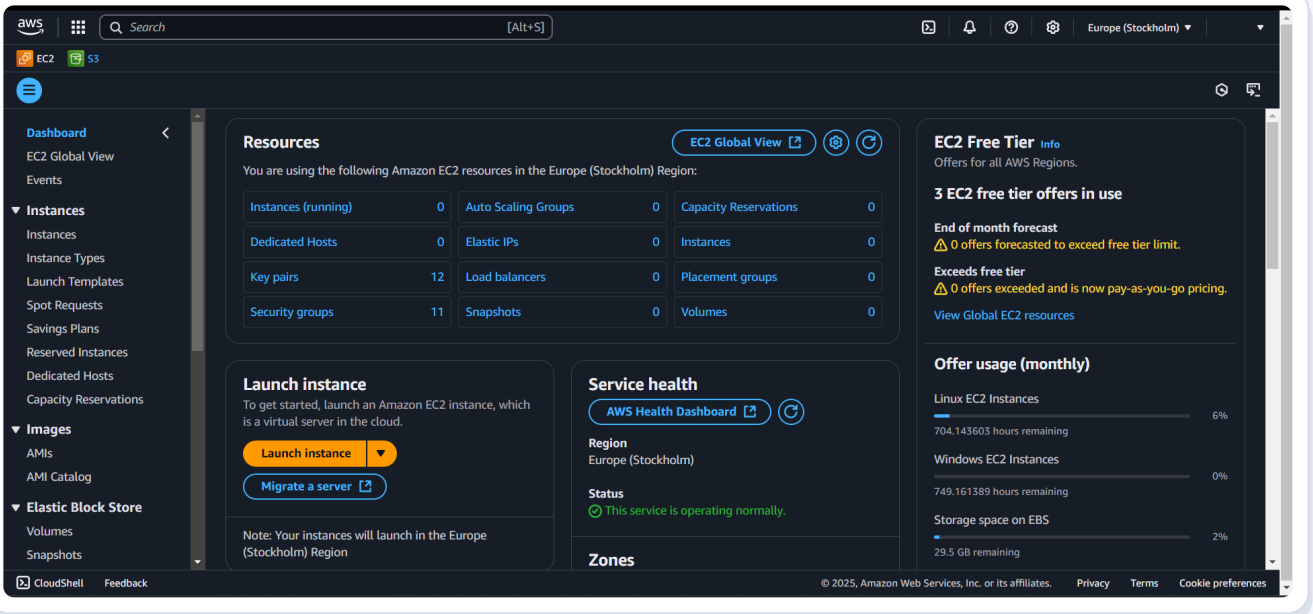
A security group acts as a virtual firewall for your instance.

01

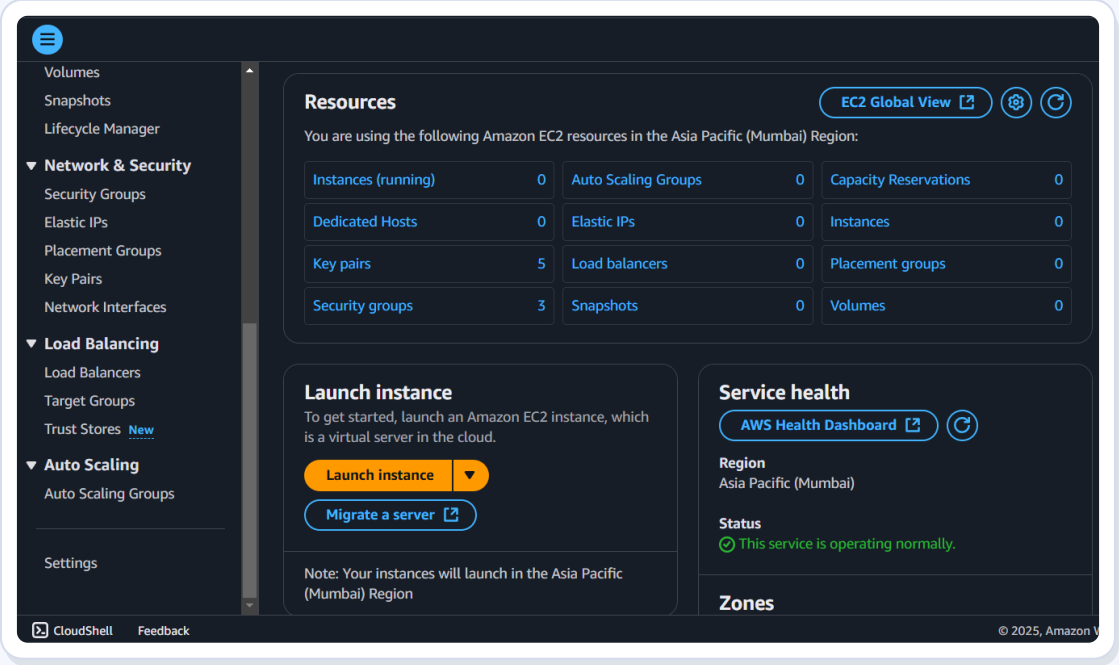
1 Open the **AWS Management Console**.



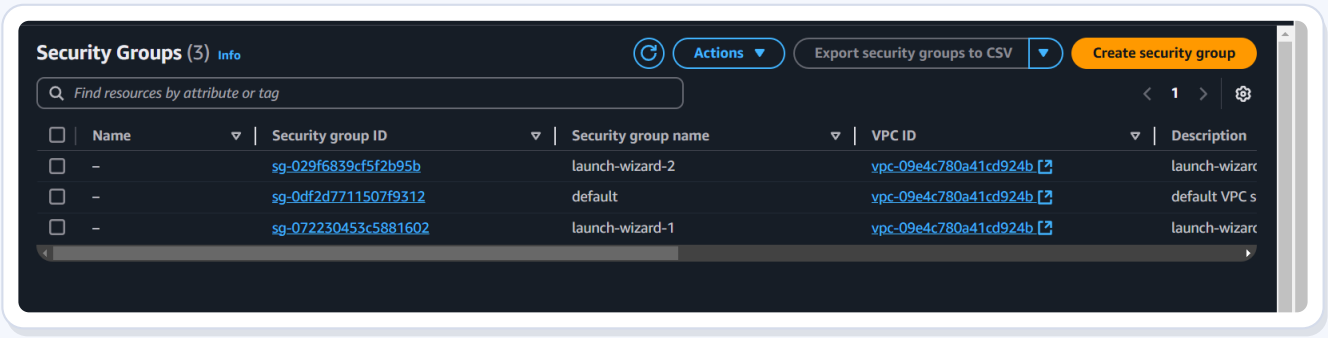
2 Search for **EC2** and navigate to the **EC2 Dashboard**.



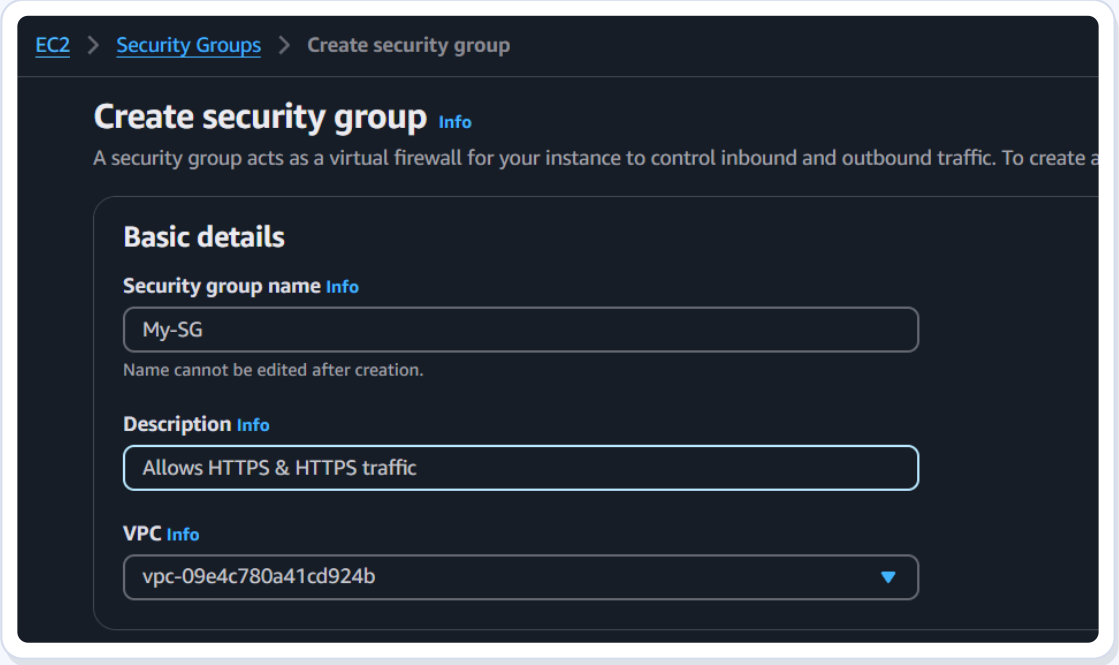
3 On the left-hand panel, find **Security Groups** and click on it.



4 Click on **Create Security Group**.



5 Enter the required details such as **name** and **description**.

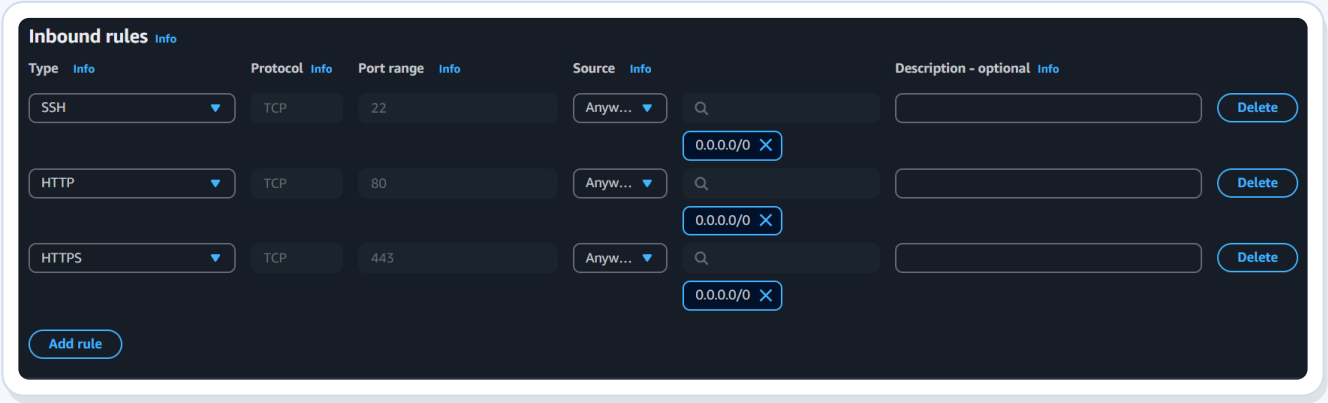


6 Under **Inbound Rules**, add the following rules and set the Source to **Anywhere-IPv4**:

SSH
22
TCP | Anywhere-IPv4 0.0.0.0/0

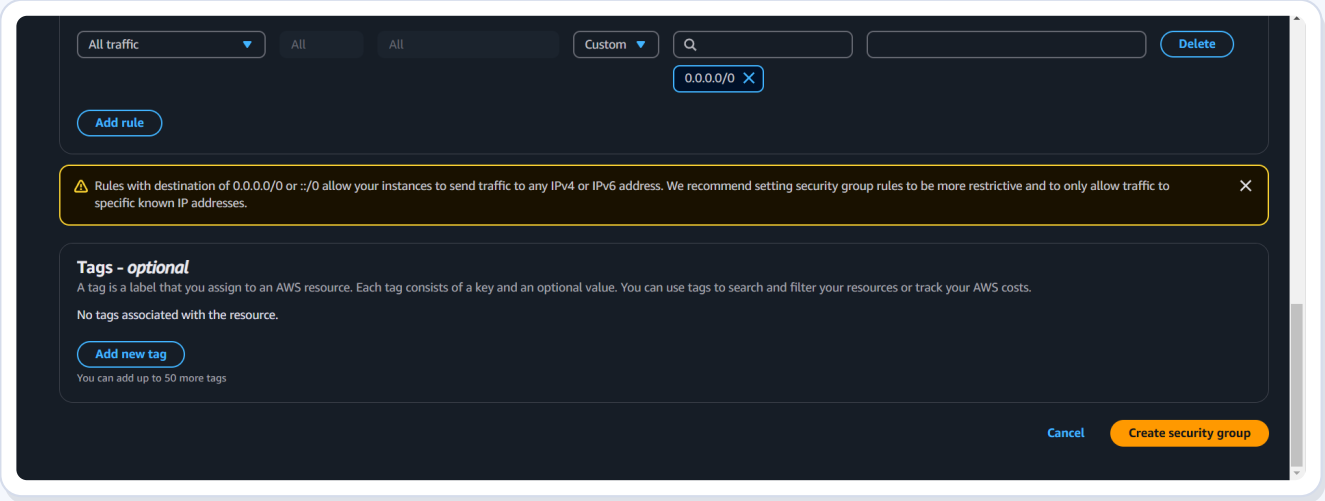
HTTP
80
TCP | Anywhere-IPv4 0.0.0.0/0

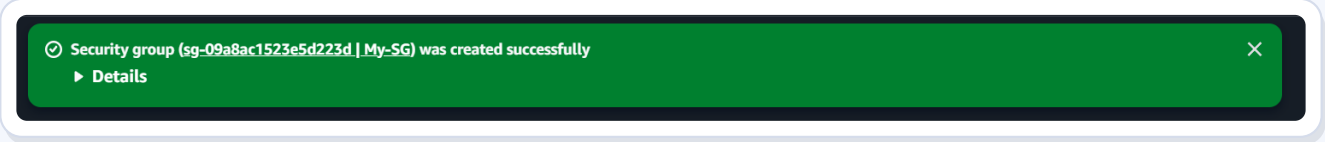
HTTPS
443
TCP | Anywhere-IPv4 0.0.0.0/0



PRO TIP Opening SSH (port 22) to 0.0.0.0/0 is fine for a short learning lab. For anything long-lived, set the SSH source to **My IP** instead.

7 Click **Create Security Group**. A green banner confirms it was created successfully.





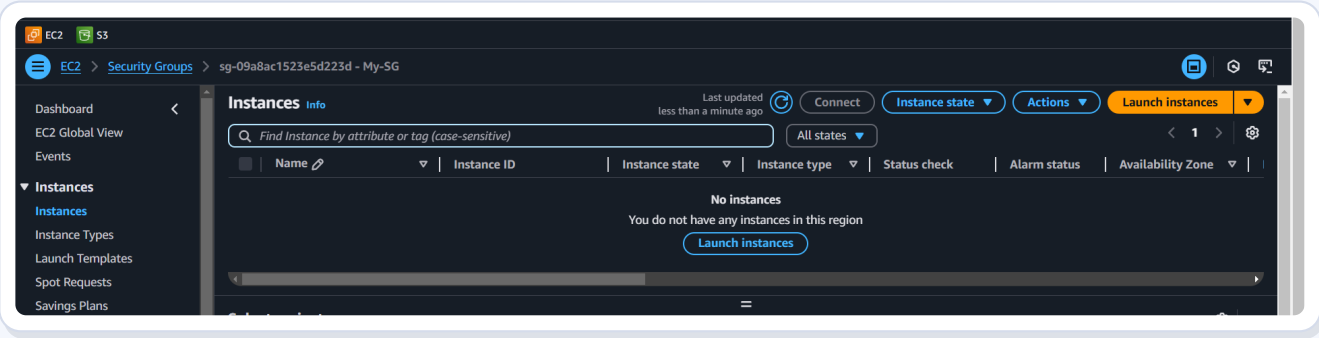
PHASE 2

Launching the EC2 Instance

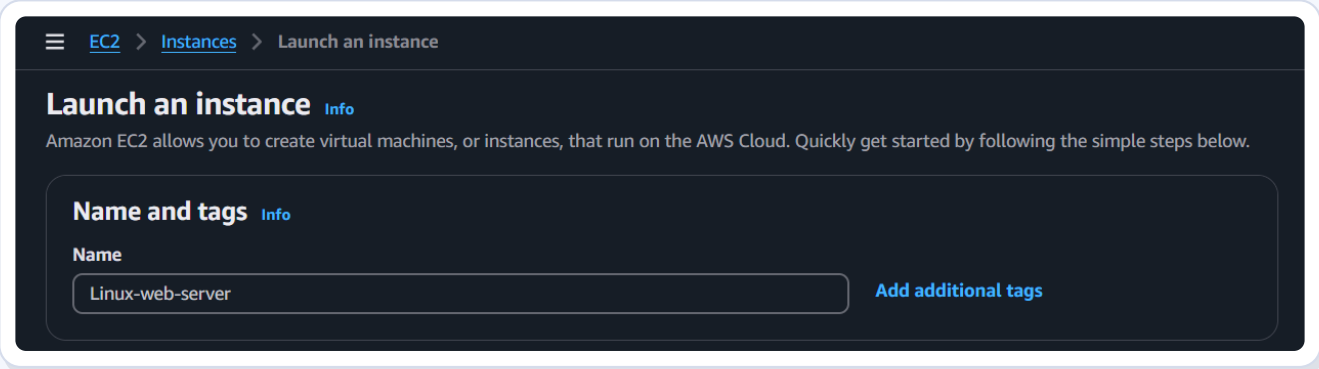
Create the virtual server that will host your website.

02

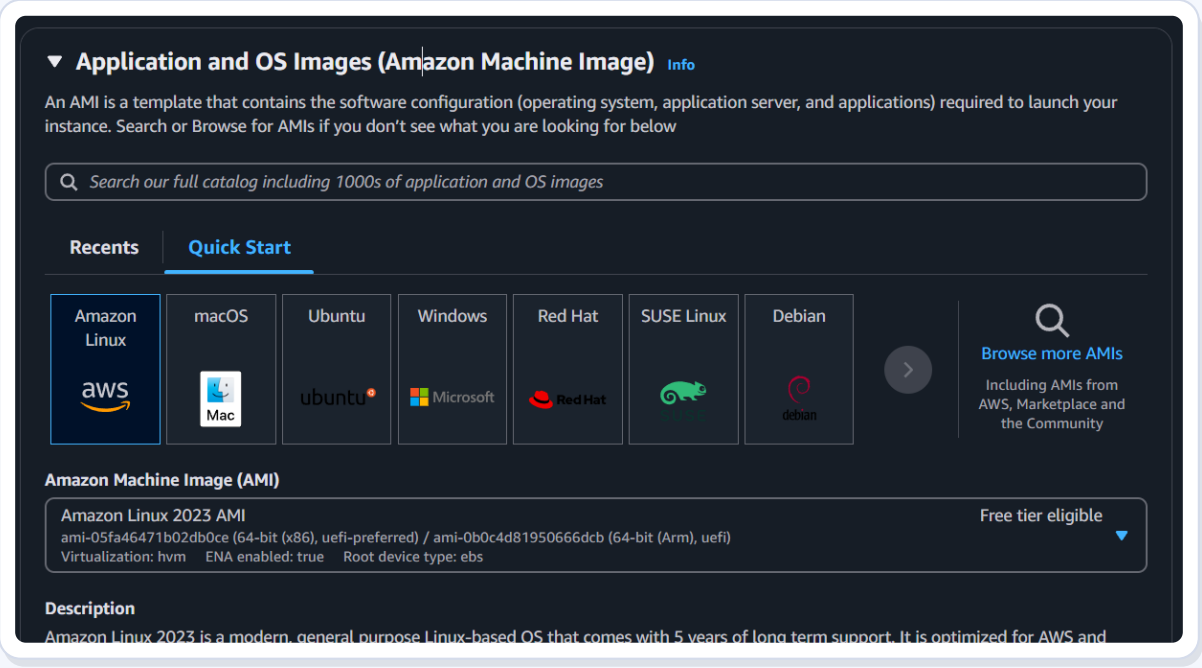
1 Go to the **EC2 Dashboard** and click on **Launch Instance**.



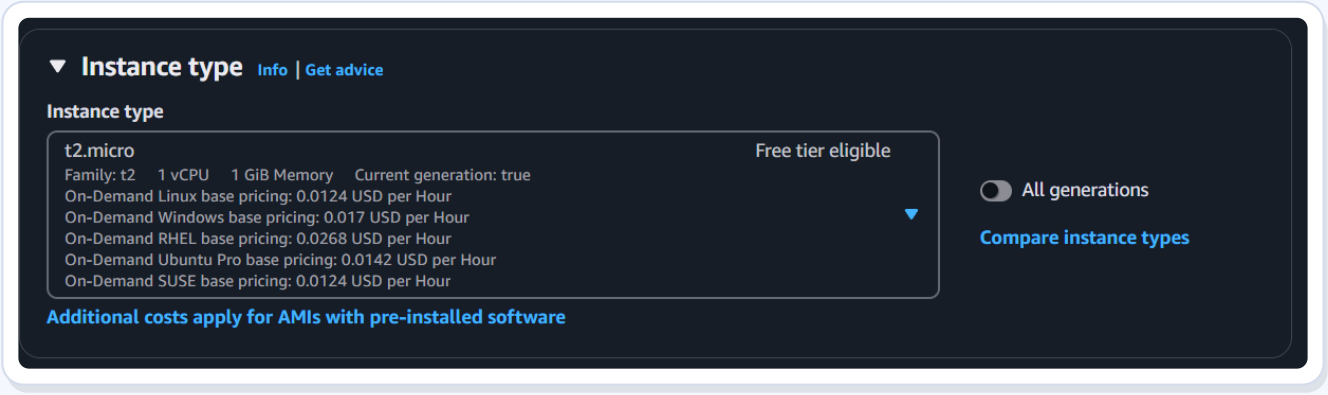
2 Enter a name for the instance, e.g., **Linux-web-server**.



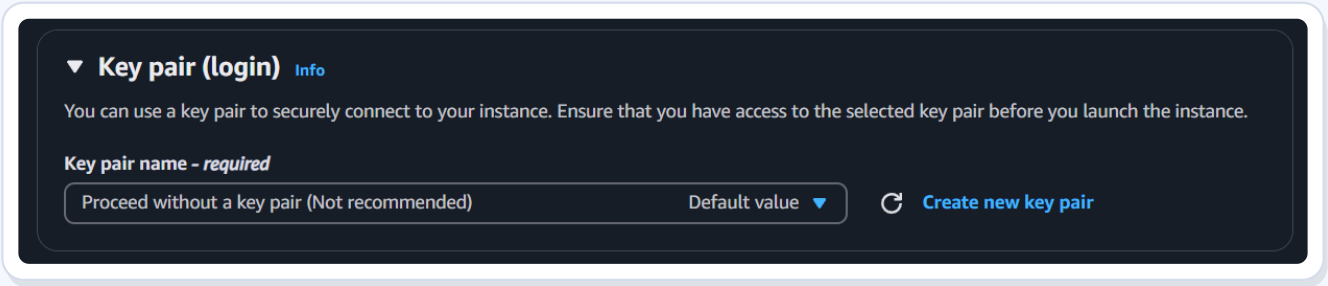
3 Keep the **Amazon Machine Image (AMI)** as **Amazon Linux**.



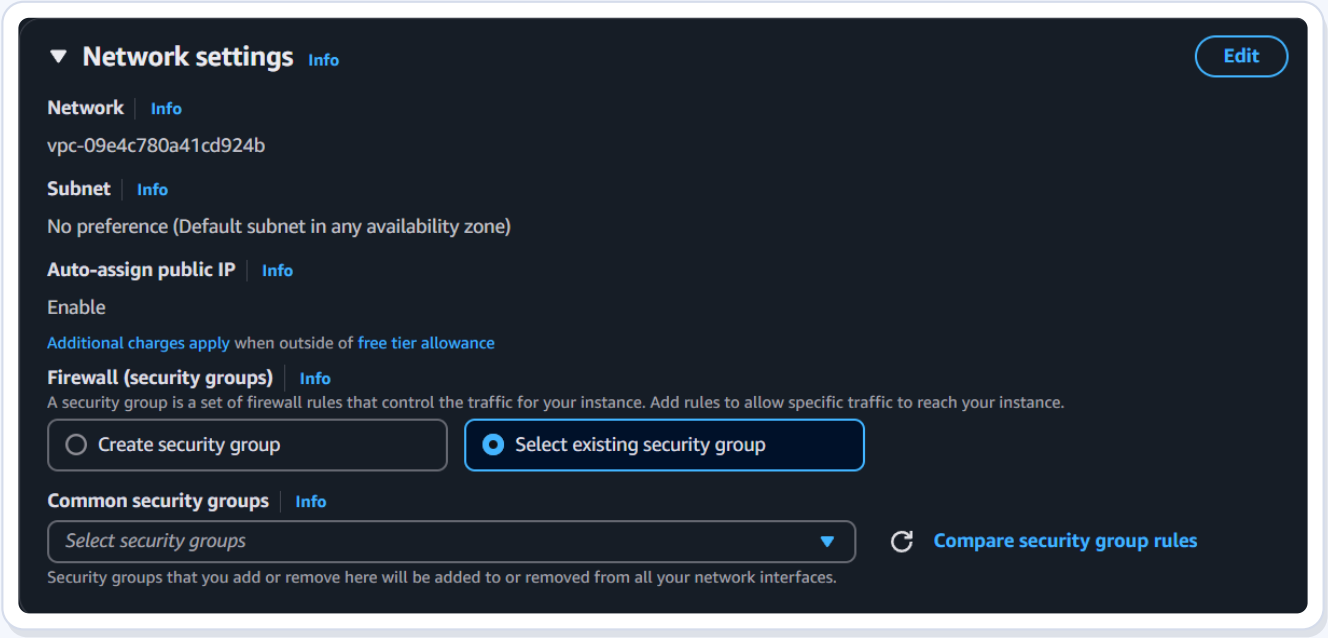
4 Select Instance Type as **t2.micro**.



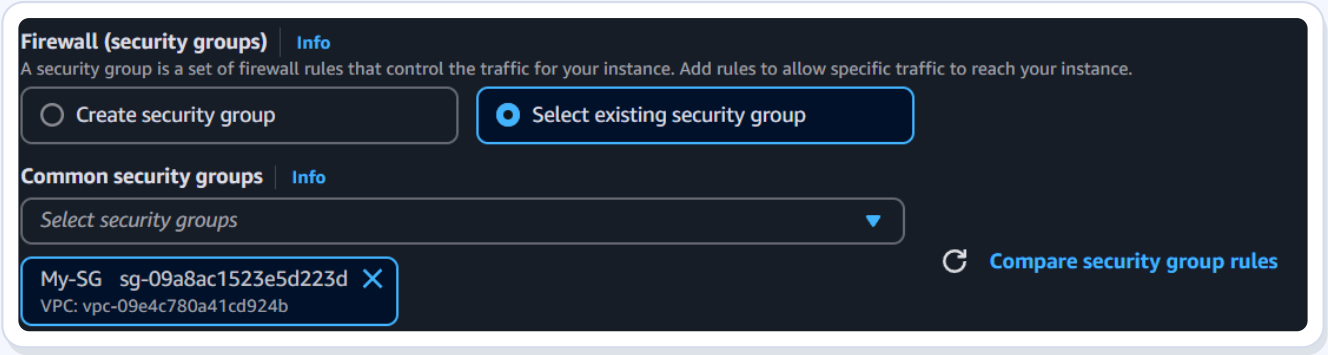
5 Proceed **without** a key pair.



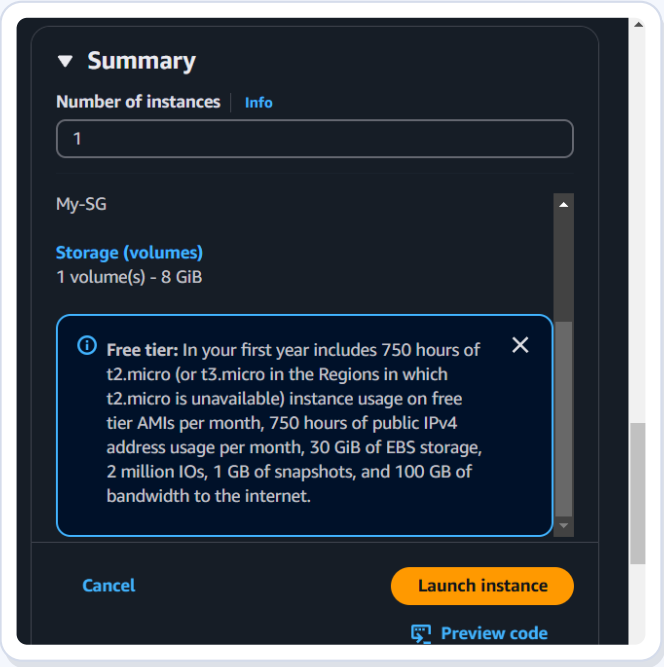
6 In **Network settings**, select **Existing Security Group** and choose the one created earlier.



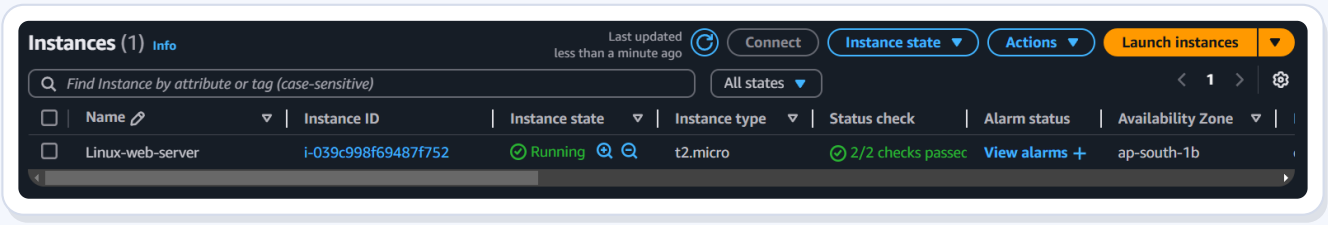




7 Keep other settings as default and click **Launch Instance**.



8 Wait until the instance passes **2/2 or 3/3 checks**.



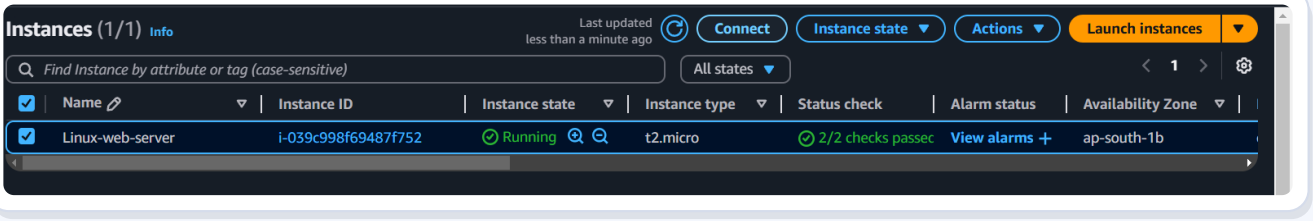
PHASE 3

Connecting to the Instance

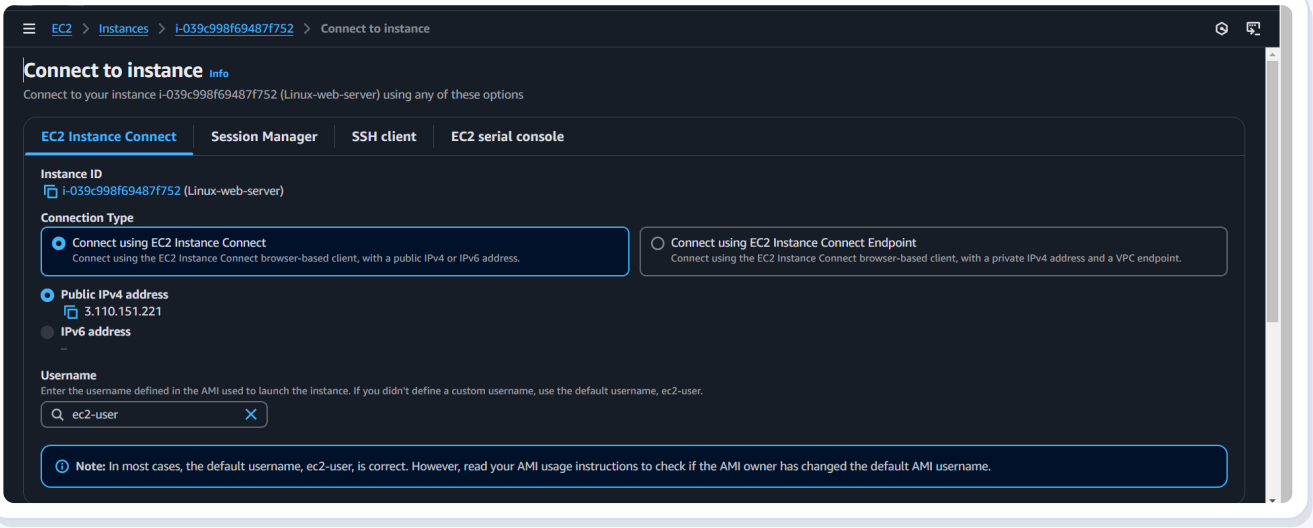
Open a browser-based terminal with EC2 Instance Connect.

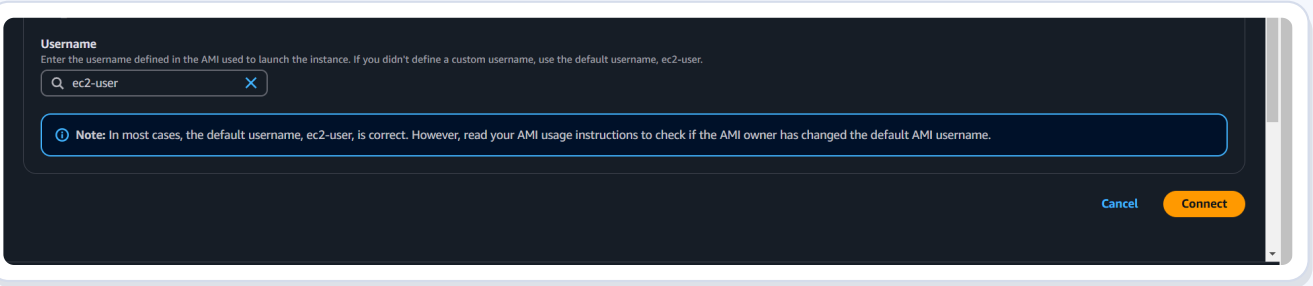
03

1 Select the **Instance** and click on **Connect**.

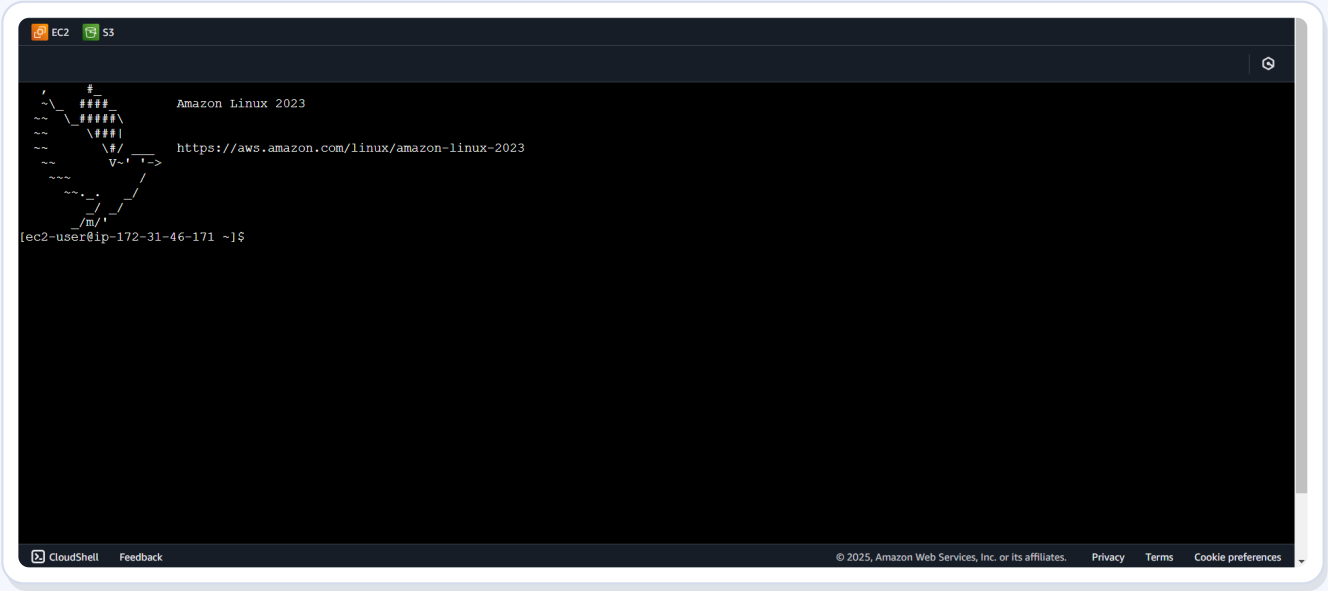


2 Scroll down to **EC2 Instance Connect** and click **Connect**.





3 Once connected, you will see a **Linux command-line interface**.



PHASE 4

Installing & Configuring Apache

Install the web server and publish a sample page.

04

1 Switch to the **root user**:

TERMINAL COMMAND

```
$ sudo su
```

What it does: Grants administrative privileges.

```
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-46-171 ~]$ sudo su
[root@ip-172-31-46-171 ec2-user]#
```

2 **Update** the system packages:

TERMINAL COMMAND

```
$ yum update -y
```

What it does: Ensures all installed packages are up to date.

```
[ec2-user@ip-172-31-46-171 ~]$ sudo su
[root@ip-172-31-46-171 ec2-user]# yum update -y
Amazon Linux 2023 Kernel Livepatch repository
Last metadata expiration check: 0:00:01 ago on Thu Jan 30 13:18:53 2025.
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-46-171 ec2-user]#
```

3 Install the Apache Web Server:

TERMINAL COMMAND

```
$ yum install httpd -y
```

What it does: Installs the Apache HTTP Server.

```
[root@ip-172-31-46-171 ec2-user]# yum install httpd -y
Last metadata expiration check: 0:00:45 ago on Thu Jan 30 13:18:53 2025.
Dependencies resolved.
================================================================================
 Package                 Architecture  Version                 Repository    Size
================================================================================
Installing:
 httpd                   x86_64        2.4.62-1.amzn2023       amazonlinux   48 k
Installing dependencies:
 apr                     x86_64        1.7.5-1.amzn2023.0.2    amazonlinux  130 k
 apr-util                x86_64        1.6.3-1.amzn2023.0.1    amazonlinux   98 k
 generic-logos-httpd     noarch        18.0.0-12.amzn2023.0.3  amazonlinux   19 k
 httpd-core              x86_64        2.4.62-1.amzn2023       amazonlinux  1.4 M
 httpd-filesystem        noarch        2.4.62-1.amzn2023       amazonlinux   14 k
 httpd-tools             x86_64        2.4.62-1.amzn2023       amazonlinux   81 k
 libbrotli               x86_64        1.0.9-4.amzn2023.0.2    amazonlinux  315 k
 mailcap                 noarch        2.1.49-3.amzn2023.0.3   amazonlinux   33 k
Installing weak dependencies:
 apr-util-openssl        x86_64        1.6.3-1.amzn2023.0.1    amazonlinux   17 k
 mod_http2               x86_64        2.0.27-1.amzn2023.0.3   amazonlinux  166 k
 mod_lua                 x86_64        2.4.62-1.amzn2023       amazonlinux   61 k

Transaction Summary
================================================================================
Install  12 Packages

Total download size: 2.3 M
Installed size: 6.9 M
Downloading Packages:
(1/12): apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64.rpm  407 kB/s |  17 kB  00:00
(2/12): apr-1.7.5-1.amzn2023.0.2.x86_64.rpm               2.7 MB/s | 130 kB  00:00
```

4 Navigate to the web directory:

TERMINAL COMMAND

```
$ cd /var/www/html
```

What it does: Changes directory to the web server's root folder.

```
Complete!
[root@ip-172-31-46-171 ec2-user]# cd /var/www/html
[root@ip-172-31-46-171 html]#
```

5 Create a sample webpage:

TERMINAL COMMAND

```
$ echo "Building Linux on AWS, where Power meets Cloud" > index.html
```

What it does: Creates an HTML file with a custom message.

```
[root@ip-172-31-46-171 ec2-user]# cd /var/www/html
[root@ip-172-31-46-171 html]# echo "Building Linux on AWS, where Power meets Cloud" >index.html
[root@ip-172-31-46-171 html]#
```

6 Start the Apache service:

TERMINAL COMMAND

```
$ service httpd start
```

What it does: Starts the Apache web server.

```
[root@ip-172-31-46-171 ec2-user]# cd /var/www/html
[root@ip-172-31-46-171 html]# echo "Building Linux on AWS, where Power meets Cloud" >index.html
[root@ip-172-31-46-171 html]# service httpd start
Redirecting to /bin/systemctl start httpd.service
[root@ip-172-31-46-171 html]#
```

7 Enable Apache to start on boot:

TERMINAL COMMAND

```
$ chkconfig httpd on
```

What it does: Ensures the web server starts automatically after a reboot.

```
Redirecting to /bin/systemctl start httpd.service
[root@ip-172-31-46-171 html]# chkconfig httpd on
Note: Forwarding request to 'systemctl enable httpd.service'.
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@ip-172-31-46-171 html]#
```

GOOD TO KNOW On Amazon Linux 2023 both commands are redirected to **systemctl**, as the terminal output above shows.

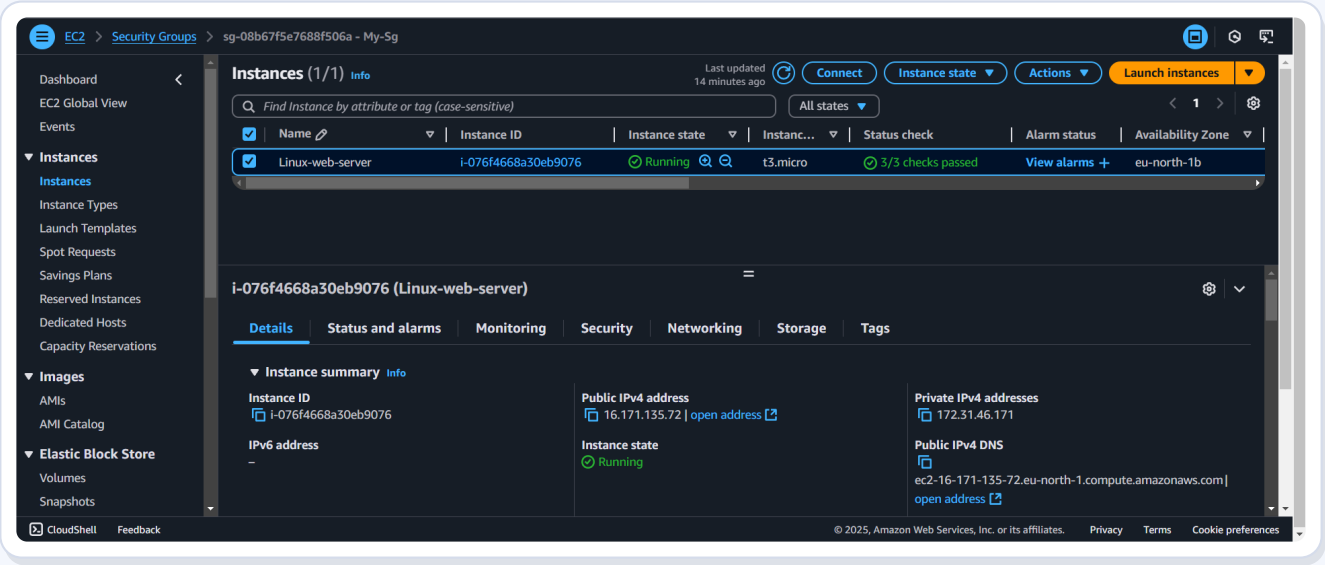
PHASE 5

Accessing the Web Page

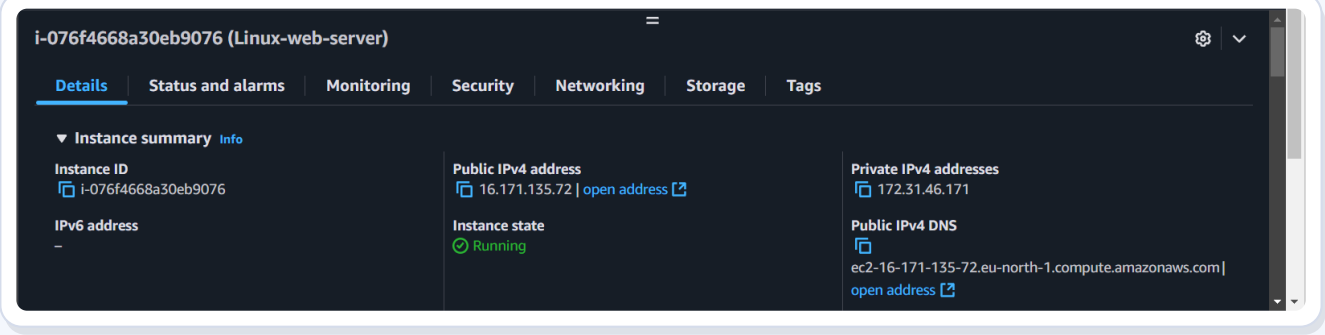
Open the public IP address in your browser.

05

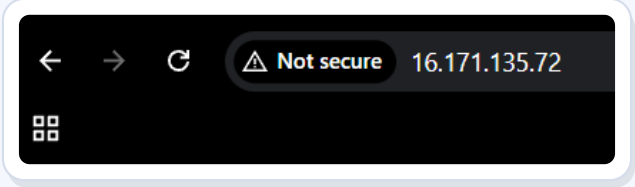
1 Go back to the **EC2 Instance Dashboard**.



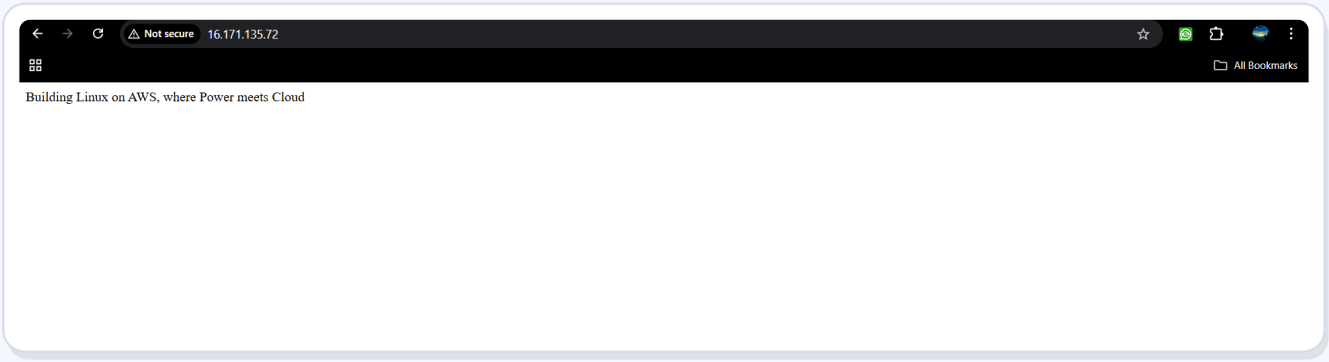
2 Copy the **Public IP Address** of the instance.



3 Paste the IP address in a **new browser tab**.



4 You should see the text *"Building Linux on AWS, where Power meets Cloud"* displayed.



SUCCESS Your Apache web server is live and serving your page to the internet.

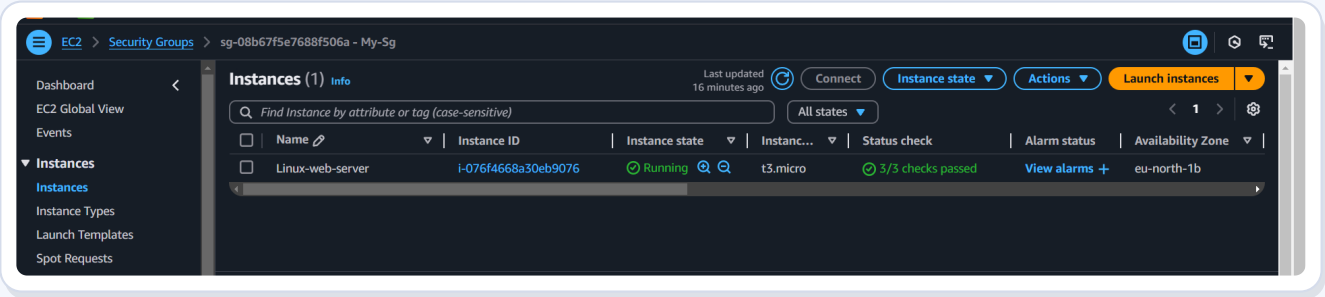
PHASE 6

Cleaning Up the Instance

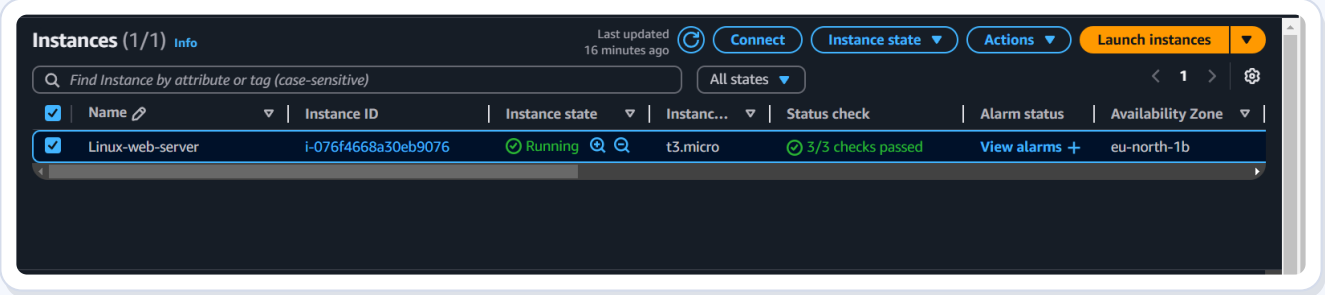
06

Terminate the instance to free up resources.

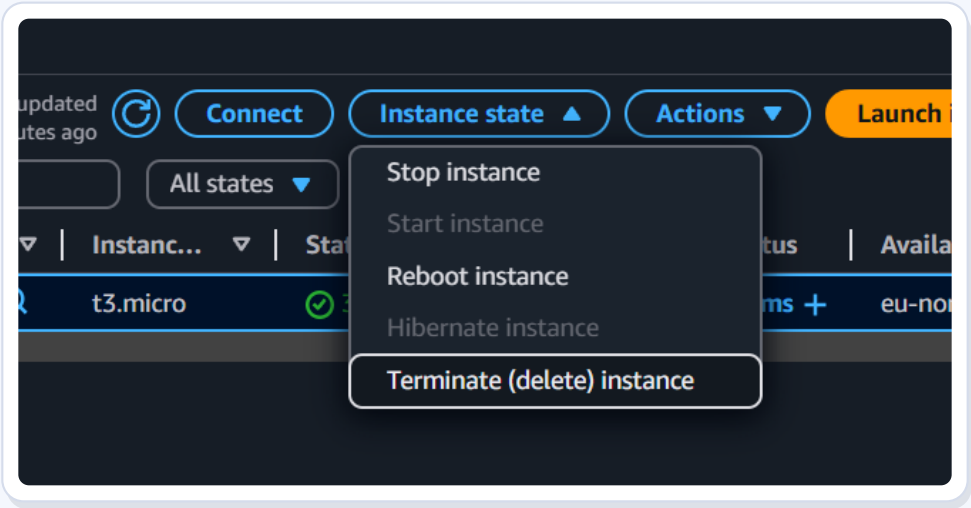
1 If you no longer need the instance, go to the **EC2 Dashboard**.



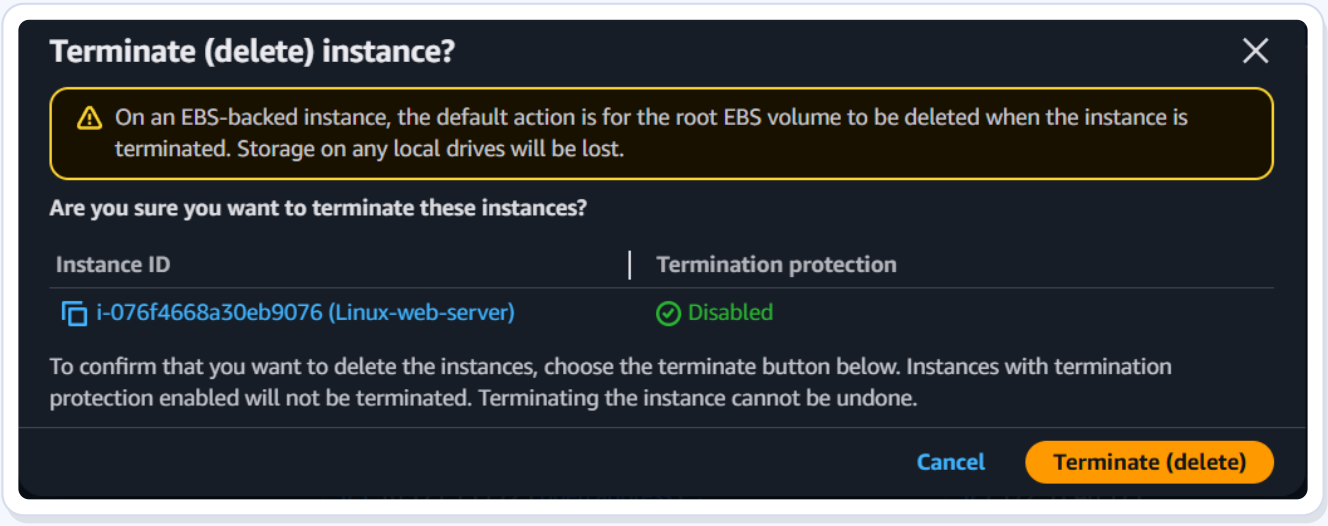
2 Select the **Instance**.

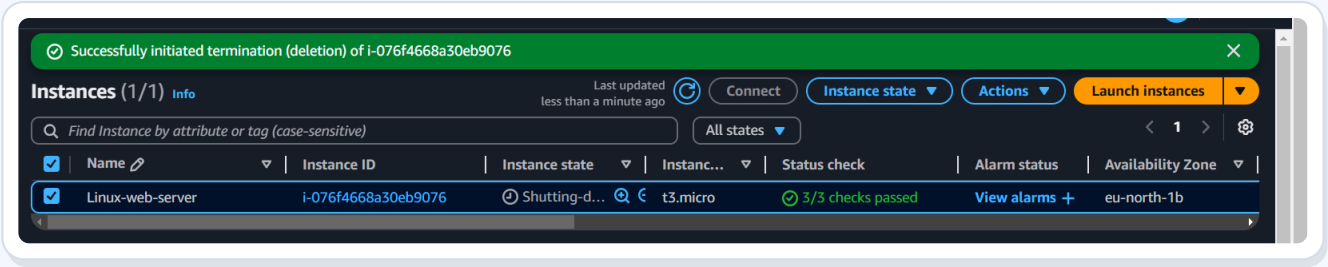


3 Click on **Terminate Instance** to delete it and free up resources.



4 Confirm by choosing **Terminate (delete)**. The state changes to *Shutting-down*.





HEADS UP Termination cannot be undone. The root EBS volume is deleted with the instance, so make sure you no longer need anything stored on it.

QUICK REFERENCE

Command Cheat Sheet

#	COMMAND	WHAT IT DOES
1	`sudo su`	Grants administrative privileges.
2	`yum update -y`	Ensures all installed packages are up to date.
3	`yum install httpd -y`	Installs the Apache HTTP Server.
4	`cd /var/www/html`	Changes directory to the web server's root folder.
5	`echo "your text" > index.html`	Creates an HTML file with a custom message.
6	`service httpd start`	Starts the Apache web server.
7	`chkconfig httpd on`	Ensures the web server starts automatically after a reboot.

PROJECT AT A GLANCE

Key Settings Used

SETTING	VALUE
Security group	My-SG, with inbound SSH (22), HTTP (80) and HTTPS (443)
Instance name	Linux-web-server
AMI	Amazon Linux 2023 AMI (free tier eligible)
Instance type	t2.micro: 1 vCPU, 1 GiB memory
Key pair	Proceed without a key pair (connect with EC2 Instance Connect)
Storage	1 volume, 8 GiB
Web server	Apache httpd, document root /var/www/html

CONCLUSION

Congratulations!

You have successfully deployed a Linux web server on AWS EC2. This setup forms the foundation for hosting websites, running applications, and exploring cloud computing. AWS EC2 provides scalability and flexibility, making it a powerful tool for developers and businesses alike.

WHAT YOU ACHIEVED

Hands-on Experience In

- **Security Groups**
  Set up AWS Security Groups
- **EC2 Instance**
  Launched and configured an instance
- **Apache Web Server**
  Installed and ran Apache
- **Web Hosting**
  Hosted a basic web page

> ***The best way to predict the future is to create it.***
>
> Peter Drucker
>
> Keep experimenting, keep building!

CREATED BY
Umesh Kumbharkar
linkedin.com/in/umesh-kumbharkar

TOPIC
AWS EC2 Web Hosting
Project Documentation